

A-level

# ENVIRONMENTAL SCIENCE

## Questions by topic: Global climate change - Set 2

### Mark scheme

AQA spec 3.2.1.2; The physical environment; textbook pp. 89-107

### Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

**Textbook references:** every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Marks |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>01.1</b> | <p><i>Any 3 from:</i></p> <ul style="list-style-type: none"> <li>• Raise river banks / build sea walls, flood barriers or wave screens</li> <li>• Permeable urban surfaces (gravel, soil) instead of concrete / tarmac to slow runoff</li> <li>• River flow management: upstream dams / low soil dams / afforestation / maintained flood plains to delay runoff</li> <li>• Raised buildings on stilts or floating houses</li> <li>• Pumping rainwater from land below river level</li> <li>• Managed retreat - abandoning land where protection costs exceed value (eg Happisburgh)</li> </ul> | 3     |
| <b>01.2</b> | <ul style="list-style-type: none"> <li>• Where the cost of flood prevention is greater than the value of what is protected, or defences would be ineffective</li> <li>• Land is abandoned as water levels rise / coast erodes; defences used only where land is very important</li> </ul>                                                                                                                                                                                                                                                                                                      | 2     |
| <b>01.3</b> | <p><i>Any 2 from:</i></p> <ul style="list-style-type: none"> <li>• Road surfaces deform as tar melts - need tarmac with a higher melting point</li> <li>• Rail track expands and buckles above the temperature it was pre-stretched for (27 °C in UK) - relay with higher pre-stress</li> <li>• Bridge damage from high river flows / debris; landslides on deforested slopes</li> </ul>                                                                                                                                                                                                       | 2     |
| <b>01.4</b> | <ul style="list-style-type: none"> <li>• CO<sub>2</sub> is captured from the flue gases (or carbon removed from the fuel before combustion)</li> <li>• Transported by pipeline / ship / tanker and stored in depleted oil or gas fields, aquifers, or used for secondary oil recovery</li> <li>• Limitation: developmental / expensive, uses energy (reducing efficiency), risk of leakage, limited storage sites</li> </ul>                                                                                                                                                                   | 3     |
|             | <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 99-100, 105-108; AQA spec 3.2.1.2.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                     |       |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Marks |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 02.1     | <p><b>Levels of response</b></p> <p><b>Level 3 (7-9 marks):</b> Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached.</p> <p><b>Level 2 (4-6 marks):</b> Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported.</p> <p><b>Level 1 (1-3 marks):</b> Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation.</p> <p><b>0 marks:</b> No relevant content.</p> <p><b>Indicative content</b> (credit other relevant, accurate points)</p> <ul style="list-style-type: none"> <li>• Control of greenhouse gases: energy conservation (transport, buildings, industry), renewable and nuclear energy replacing fossil fuels, CCS, reduced deforestation and afforestation (carbon sequestration), methane capture from landfill / digesters, HFC controls, changes in agriculture</li> <li>• Geoengineering: white roofs, ocean fertilisation, solar shades - largely untried, possible unpredictable consequences</li> <li>• International agreements: Kyoto Protocol, Paris Agreement - targets, but non-binding / uneven participation; carbon trading and taxes; comparison of per capita carbon footprints between countries</li> <li>• Evidence of limited success: global CO<sub>2</sub> concentrations still rising (about 280 ppm pre-industrial to over 410 ppm); temperatures up about 1 °C; emissions in some developed countries falling but rising in developing economies</li> <li>• Reasons for difficulty: fossil fuels underpin economies, need for major lifestyle change, cost, time delays and natural fluctuations masking effects, political disagreement, developing countries' right to develop</li> <li>• Adaptation strategies (flood defences, drainage, managed retreat, infrastructure changes) reduce impacts but do not address the cause</li> <li>• Comparison with the Montreal Protocol's success for ozone</li> <li>• Judgement: partial success - technology exists and some regional reductions achieved, but global strategies have not yet stopped the rise; tipping points make urgency greater</li> </ul> | 9     |
| 02.2     | <ul style="list-style-type: none"> <li>• The total greenhouse gas / CO<sub>2</sub> emissions caused by an individual, organisation, product or country (often expressed per capita per year)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 89-107; AQA spec 3.2.1.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |
| 03.1     | <ul style="list-style-type: none"> <li>• 2020 has higher peak flow in the wet season (32 vs 22 m<sup>3</sup> s<sup>-1</sup> in March)</li> <li>• 2020 has much lower dry-season flow (2 vs 9 m<sup>3</sup> s<sup>-1</sup> in July / August)</li> <li>• Flow is much more variable / seasonal in 2020; the river almost dries up</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 3     |
| 03.2     | <ul style="list-style-type: none"> <li>• In 1990 snow / ice stored wet-season precipitation and released it gradually as meltwater during the dry season, evening out the flow</li> <li>• With less ice, precipitation now falls as rain / melts immediately and runs off quickly, increasing wet-season flow and flood risk</li> <li>• Little stored ice remains to melt in the dry season so flow falls very low</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 3     |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                              | Marks |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>03.3</b> | <i>Any 2 from:</i> <ul style="list-style-type: none"> <li>• Water shortages for irrigation / drinking / HEP in the dry season</li> <li>• Flooding in the wet season</li> <li>• Sudden floods from ice-lake outbursts (glacial lake outburst floods)</li> </ul>                                                                                                | 2     |
| <b>03.4</b> | <ul style="list-style-type: none"> <li>• Surface meltwater flows down cracks to the base, lubricating the ice so it slides faster; if this exceeds the faster melting at the front, the front advances</li> <li>• But if snowfall at the source has not increased, total volume falls; both retreating and extending glaciers can indicate warming</li> </ul> | 2     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 94-96; AQA spec 3.2.1.2.                                                                                                                                                                                                                                         |       |